

Roman Ibrahimov

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EDUCATION AND PROFESSIONAL DEVELOPMENT	University of California, Berkeley, CA, USA M.S. , Mechanical Engineering Purdue University, West Lafayette, IN, USA M.S. , Aeronautics and Astronautics • IEEE RAS Summer School on Multi-Robot Systems, CTU, Prague, Czechia, 1-5 Aug 2022 Multi-UAV control, perception, localization, and planning • DroneCamp, University of California, ANR, Monterey, California, USA, 27 June - 1 July, 2022 UAV hardware & sensors, mission planning, flight skills, safety & regulations ADA University, Baku, Azerbaijan B.S. Summa Cum Laude , IT and Systems Engineering • ITMO University, St. Petersburg, Russia, <i>Spring 2017</i> Exchange Student, Control Systems and Robotics • Middle East Technical University, Ankara, Turkey, <i>Spring 2016</i> Exchange Student, Electrical and Electronics Engineering	<i>Aug 2023–May 2025</i> <i>Sep 2014–May 2018</i>
RESEARCH INTERESTS	Robot Learning; Contact-Rich Manipulation; Sim-to-Real Transfer; Control & Dynamics; Robotic Hardware Design	
PUBLICATIONS	R. Ibrahimov, S. Mozaffari, and A. Adel, “Contact-Rich Manipulation in Construction via Zero-Shot Learning: A Diffusion Policy-Guided Adaptive Control,” <i>Automation in Construction</i> (submitted), 2026. R. Ibrahimov, S. Mozaffari, and A. Adel, “Sensing Modalities for Robust Sim-to-Real Diffusion Policies in Contact-Rich Manipulation,” <i>in preparation</i>, 2026. R. Ibrahimov, T. Yang, and M. W. Müller, “Kalman Filter-Based Drift Detection and Mitigation of Visual-Inertial Odometry in UAVs,” <i>American Control Conference (ACC)</i>, 2025. T. Yang, R. Ibrahimov, and M. W. Müller, “Towards Safe and Efficient Through-the-Canopy Autonomous Fruit Counting with UAVs,” <i>IEEE International Conference on Robotics and Automation (ICRA)</i>, 2025. R. Ibrahimov and J. Heinen, “Multi-Robot Distributed Optimization for Exploration and Mapping of Unknown Environments using Bioinspired Tactile-Sensor,” <i>arXiv preprint</i>, 2025. A. Behjat, R. Ibrahimov, A. Lenjani, A. Barket, K. Martinus, A. Maghareh, D. Whitaker, I. Bilonis, and S. Dyke, “A Computational Framework for the Evaluation of Resilience in Deep Space Habitat Systems,” <i>ASME 2022 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference</i>, St. Louis, Missouri, USA, Aug. 14-17, 2022. E. Tsykunov, A. Fedoseev, E. Dorzhieva, R. Agishev, R. Ibrahimov, D. Vasquez, L. Labazanova, and D. Tsetserukou, “DroneStick: Flying Joystick as a Novel Type of Interface,” <i>ACM SIGGRAPH Asia 2021 Emerging Technologies</i>, Virtual, Dec. 14-17, 2021. E. Karmanova, V. Serpiva, S. Perminov, R. Ibrahimov, A. Fedoseev, and D. Tsetserukou, “SwarmPlay: A Swarm of Nano-Quadcopters Playing Tic-tac-toe Board Game against a Human,” <i>ACM SIGGRAPH 2021 Emerging Technologies</i>, Virtual, Aug. 9-13, 2021.	

R. Ibrahimov, N. Zherdev, and D. Tsetserukou, “DroneLight: Drone Draws in the Air using Long Exposure Light Painting and ML,” 29th IEEE International Conference on Robot & Human Interactive Communication (IEEE RO-MAN 2020), Naples, Italy, Aug. 31-Sept. 4, 2020.

E. Tsykunov, R. Agishev, **R. Ibrahimov**, T. Moriyama, L. Labazanova, H. Kajimoto, and D. Tsetserukou “SwarmCloak: Landing of Two Micro-Quadrotors on Human Hands Using Wearable Tactile Interface Driven by Light Intensity,” 2020 IEEE Haptics Symposium, Washington DC, USA, March 28-31, 2020.

R. Ibrahimov, E. Tsykunov, V. Shirokun, A. Somov, and D. Tsetserukou, “DronePick: Object Picking and Delivery Teleoperation with a Drone Controlled by a Tactile Wearable,” 28th IEEE International Conference on Robot & Human Interactive Communication (IEEE RO-MAN 2019), New Delhi, India, 2019.

E. Tsykunov*, **R. Ibrahimov***, D. Vasquez, D. Tsetserukou, “SlingDrone: System for Navigation and Interaction with the Environment Using a Single Drone and VR,” 25th ACM Symposium on Virtual Reality Software and Technology (VRST 2019), Sydney, Australia, 2019.

*- authors contributed equally to the paper.

E. Tsykunov, R. Agishev, **R. Ibrahimov**, L. Labazanova, T. Moriyama, H. Kajimoto, D. Tsetserukou, “SwarmCloak: Landing of a Swarm of Nano-Quadrotors on Human Arms,” Int. Conf. on Computer Graphics and Interactive Technologies (ACM SIGGRAPH Asia 2019), Emerging Technologies, Brisbane, Australia, 2019.

E. Tsykunov, R. Agishev, **R. Ibrahimov**, L. Labazanova, A. Tleugazy, and D. Tsetserukou, “SwarmTouch: Guiding Swarm of Nano-Quadrotors with Impedance Control using Wearable Tactile Interface,” IEEE Transactions on Haptics, 2019.

G. Yashin, D. Trinitatova, R. Agishev, **R. Ibrahimov**, and D. Tsetserukou, “AeroVR: Virtual Reality Teleoperation System for the UAV Robotic Manipulator,” 19th IEEE International Conference on Advanced Robotics (ICAR 2019), Belo Horizonte, Brazil, 2019.

PROJECT EXPERIENCE

Contact-Rich Robotic Manipulation for Construction, Princeton University *July 2025–present*

- Developing a contact-rich manipulation framework for automated assembly tasks using a UR20 robot.
- Generating large-scale synthetic demonstration datasets in IsaacSim for robust sim-to-real transfer.
- Training and evaluating diffusion policy augmented with adaptive control for zero-shot real-world performance.

Autonomous Drone Inspection of Caltrans Bridges, UC Berkeley *Aug 2023–May 2025*

- Implemented visual-inertial odometry with OpenVINS, integrating IMU and stereo camera.
- Configured and tested hardware, including RealSense D455 for depth imaging, Qualcomm RB5 for onboard computing and Pixracer R15 for flight control.
- Managed system communication via the Robot Operating System (ROS).

An inspection robot for Duke Energy Power Lines, Purdue University *May 2022–Aug 2023*

- Implemented Model Predictive Control (MPC) for trajectory tracking along power lines.
- Developed line detection algorithms using OpenCV for enhanced visual recognition.
- Modeled and simulated robot dynamics.
- Configured and tested Argo rover for optimal field deployment and reliability.

Resilient Extraterrestrial Habitats, NASA RETH Institute, Purdue University *May 2021–Aug 2023*

- Designed a control-theoretic autonomy framework to support resilient design and operation.
- Built automated active learning framework with robots and humans-in-the-loop.
- Developed methods for detection and diagnosis of anticipated and unanticipated faults.
- Contributed to establishing SmartHabs with autonomous abilities to sense, anticipate and respond.

Bio-inspired nano-quadcopter for map building, Purdue University *Sep 2021–May 2022*

- Developed an API on FreeRTOS to read sensor reading on the quadcopter.
- Transmitted sensor data to the ROS base station via radio communication.
- Collected and processed point cloud data to construct detailed environmental maps.
- Applied machine learning techniques to predict the map of the environment.

Human-Drone Interaction through a Tactile Wearable, Skoltech *Oct 2018–May 2020*

- Implemented human-drone communication using impedance control and vibrotactile feedback.
- Designed a tactile wearable equipped with eccentric rotating mass (ERM) motors for intuitive interaction.

- Created a Virtual Reality (VR) application in C# to support teleoperation and aerial manipulation tasks.
- Developed remote object manipulation techniques using drones.

Balloon Satellite for Testing Solar Cells in High Altitude, Skoltech

Sep 2018–Jan 2019

- Applied Model-based Systems Engineering principles to ensure mission success.
- Constructed a CubeSat featuring an onboard controller, solar cells, storage devices, and GPS tracking.
- Gathered and analyzed current, voltage, and temperature (CVT) data from solar cell tests.
- Retrieved payload with no damage after landing from 35km maximum altitude.

Gas Leak Detecting Mobile Robot for NICA Collider, JINR

Summer 2018

- Designed autonomous navigation system around elliptical collider.
- Built a mobile robot with an on-board temperature camera.
- Developed Computer Vision (CV) algorithm based on Python for detection nitrogen gas leak from the collider.
- Came up with computer-based user interface for remote monitoring.

WORK EXPERIENCE

Robotics Research Associate, Princeton University, NJ, USA

Jul 2025–present

- Develop contact-rich robotic manipulation strategies for construction assembly using UR20 manipulators.
- Generate large-scale synthetic datasets in Isaac Sim for sim-to-real transfer.
- Implement ROS 2 control pipelines with force/torque feedback for contact-rich manipulation.
- Collaborate with faculty and students on experimental design, system integration, and publications.

Research Co-Op, MaxInsights, Inc., San Jose, CA, USA

Jan 2025–May 2025

- Built a teleoperation pipeline integrating DexCap motion capture with Rokoko gloves on a Realman manipulator for dexterous manipulation data collection.
- Developed real-time data streaming and coordinate-frame calibration between the glove, motion capture system, and robot end-effector.
- Contributed to data collection workflows for training embodied AI policies on dexterous manipulation tasks.
- Demonstrated the full pipeline live at IEEE ICRA 2025.

Graduate Student Instructor, UC Berkeley, CA, USA

Jan 2025–May 2025

- ENGIN07 Computer Programming for Scientists and Engineers.
- Led coding labs and exercises, supporting course delivery.
- Conducted hands-on sessions to apply concepts in programming tasks.
- Held office hours to answer questions and assist with assignments.
- Designed and graded coding assignments, providing feedback on student work.

Graduate Research Assistant, UC Berkeley, CA, USA

Aug 2023–May 2025

- Designed and tested control algorithms for UAV stability and obstacle avoidance in complex environments.
- Optimized data processing workflows, enhancing efficiency and accuracy for real-time inspection feedback.
- Collaborated with interdisciplinary teams to troubleshoot hardware and software challenges.
- Presented research findings within lab meetings and contributed to technical publications on UAV autonomy.

Graduate Research Assistant, NASA RETH Institute, IN, USA

May 2021–Aug 2023

- Developed real-time fault detection algorithms and situational awareness systems to enhance safety and reliability in extreme environments.
- Designed and configured hardware systems, including specialized sensors, actuators, and computational units, to support autonomous operation in space habitats.
- Conducted simulations and tests to validate habitat resilience against environmental and operational disruptions.
- Collaborated with interdisciplinary teams to integrate control-theoretic approaches into habitat design, advancing NASA's SmartHab initiative with enhanced autonomous capabilities.

Teaching Assistant, Purdue University

Jan 2021–May 2022

- CNIT 155 Introduction to Software Development Concepts (in Python).
- Supported course delivery by leading coding labs and exercises, guiding students in foundational Python programming and problem-solving.
- Conducted hands-on lab sessions, helping students implement concepts through coding exercises and real-world applications.
- Held office hours to address individual student questions, clarify complex topics, and provide assignment

assistance.

- Created and evaluated coding assignments, offering constructive feedback to enhance students' programming skills and understanding of software development principles.

Intern, Universal Robots, Moscow, Russia

Summer 2019

- Assisted in research and development for Human-Robot Collaboration (HRC) applications involving quadcopters to enhance interaction and safety.
- Participated in designing and testing control strategies to improve quadcopter responsiveness and collaboration capabilities.
- Supported hardware configuration and troubleshooting to ensure seamless integration of HRC functionalities.
- Documented findings and presented insights to the team, contributing to advancements in HRC applications.

Intern, Azercosmos OJSCo, Baku, Azerbaijan

Summer 2017

- Supported the Ground Control Department in monitoring and maintaining satellite communication systems.
 - Assisted with network configuration and troubleshooting to ensure stable connectivity and optimal system performance.
 - Gained experience in analyzing and managing data flow within ground control networks.
 - Documented operational procedures and reported on system performance, contributing to improved workflow efficiency.
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HONORS AND AWARDS

American Control Conference Full Travel Grant \$2000

April 2025

Purdue University Poster Symposium Best Poster Award \$750

April 2022

Skolkovo Foundation Full Graduate Scholarship

2018-2020

ACM SIGGRAPH Asia Best Demonstration Award

November 2019

Skolkovo Presidential Scholarship

November 2019

Skolkovo Presidential Travel Award, \$5000

November 2019

Skolkovo Presidential Travel Award, \$4500

November 2019

President's List of Honor, ADA University

2014-2017

Erasmus+ Exchange Scholarship, METU

Spring 2016

FameLab World Finalist, Cheltenham, UK

June 2015

1st place, Microsoft ImagineCup Competition, Baku, Azerbaijan

May 2015

4th place, International Rudolf Ortvy Competition in Physics, Hungary

Dec 2014

MENTORING EXPERIENCE

Princeton University:

- Joshua Song, BS in Electrical and Computer Engineering
- Moyu Yamaguchi, BS in Civil and Environmental Engineering

UC Berkeley:

- Jannik Matthias Heinen, MEng in Mechanical Engineering (PhD student at Aalto University, Finland)
- Milan Rosic, MEng in Aerospace Engineering (safety engineer at Lucid Motors)
- Yuxuan Peng, MEng in Mechanical Engineering
- Ruonan Yang, MEng in Mechanical Engineering

Purdue University:

- Anderson Xu, BS in Aero/Astro (PhD student at Purdue University)
 - Benjamin Krugman, BS in Mechanical Engineering
 - Fahad Tajiki, BS in Electrical Engineering (firmware engineer at Saab, Inc.)
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REVIEWER FOR CONFERENCE AND JOURNALS

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026, IEEE Robotics and Automation Letters (RA-L) 2024, RA-L 2023, RA-L 2022, IEEE International Conference on Robotics and

TECHNICAL
SKILLS

Programming: C/C++, Java, Python, MATLAB/Simulink, LabVIEW, $\LaTeX$

Robotics: Robot Operating System (ROS), ROS2, Gazebo, NVIDIA Isaac Sim, RViz, Unity

MCUs: Qualcomm RB5, myRIO, Arduino, Libelium

Mechanical: CAD (SolidWorks), 3D Printing, soldering, laser/plasma cutting